

Thummala Pramod Kumar

📍 Visakhapatnam, Andhra Pradesh 📞 +91 9347667036 ✉ pramodlucky2525@gmail.com
🌐 [linkedin.com/in/Pramod-kumar](https://www.linkedin.com/in/Pramod-kumar) 🐙 github.com/Pramod-kumar-25

SUMMARY

AI Engineer specializing in Agentic AI systems, multi-agent architectures, and production-grade RAG pipelines. Proficient in Python, FastAPI, vector databases (pgvector), and Cloud AI services (OpenAI API, Azure OpenAI), with hands-on experience building stateful AI workflows, agent planning, tool use, and human-in-the-loop execution systems. Proven ability to architect scalable, reliable AI platforms across backend, retrieval, and orchestration, seeking to contribute to a product-driven engineering team building real-world Agentic AI solutions.

EDUCATION

Vignan's Institute of Information Technology (VIIT), Visakhapatnam, India	Nov 2021 – June 2024
B.Tech in Artificial Intelligence and Data Science	CGPA: 7.9/10
Govt. Polytechnic College, Srikakulam, India	July 2018 – Sept 2021
Diploma in Computer Science and Engineering	CGPA: 8.7/10

PROFESSIONAL EXPERIENCE

Agentic AI Engineer	August 2025 – Present
<i>OTTOBON Prof Svc Pvt. Ltd, India</i> 📄	

AI Digital Twin for Experts

- Architected and built a configuration-driven Digital Twin platform in Python to model expert workflows, developing **Multi-agent systems** capable of **Planning**, **Tool use**, and stateful AI orchestration.
- Designed a zero-trust execution engine with FastAPI and LangGraph-style state machine patterns for adaptive data gathering, context retrieval, safety-rule evaluation, and agent coordination.
- Implemented a hybrid Retrieval-Augmented Generation (RAG) pipeline combining vector search, lexical retrieval, confidence gating, and hierarchical knowledge hydration over pgvector/Supabase knowledge stores.
- Built structured knowledge ingestion pipelines that transformed expert documentation into retrieval chunks, embeddings, audit records, and synchronized Obsidian knowledge artifacts.
- Developed pluggable orchestration components for extractors, safety rules, embedding services, repositories, and export layers, improving modularity and maintainability.
- Engineered human-intercept and AI safety guardrail mechanisms for high-stakes workflows, including anomaly detection, escalation protocols, execution trace logging, and review-driven intervention.
- Created asynchronous workflow support using saga-style coordination, background synthesis jobs, and durable session transitions for booking and follow-up flows.
- Delivered a React-based control plane and patient/doctor workflow interfaces for configuration management, pre-consultation intake, escalation review, audit visibility, and appointment operations.

AI Learning Platform (Ottobon Academy)

- Architected a production-grade Retrieval-Augmented Generation (RAG) pipeline in Python using OpenAI embeddings, pgvector, and LangChain for content chunking, hybrid vector search, memory-aware summarization, and usage quota enforcement.
- Built a persistent, conversational AI tutor system in Python with conversation memory, learner grounding, and real-time response personalization via async FastAPI services.
- Led end-to-end development of an AI-native learning platform using a Python-first stack (FastAPI) for core logic and selective Node.js for real-time features; deployed on AWS and launched multiple learner cohorts.
- Designed and implemented backend services in Python for course orchestration, session management, dynamic module delivery, and mastery-based learner progression.
- Integrated secure Google OAuth2, JWT access/refresh rotation, and role-based access control across Python (FastAPI) and Node.js microservices.
- Implemented structured learning guardrails including quiz-gated unlocks, cooldown timers, and prerequisite validation and shipped progress analytics dashboards to enable instructor intervention.
- Used Node.js (Express) selectively for WebSocket-based activity feeds, while migrating AI-intensive workloads to Python for scalability, maintainability, and operational clarity.

Graduate Technical Apprentice (IT)

Dredging Corporation of India Limited, Visakhapatnam, India

September 2024 – July 2025

- Supported enterprise operations by maintaining MS Dynamics 365 (ERP) modules for payroll and HR processes.
- Delivered frontline IT support, resolving hardware, software, and network issues for 100+ employees.
- Created interactive Power BI dashboards to visualize operational data, aiding departmental decision-making.

TECHNICAL SKILLS

AI & GenAI: Agentic AI, Multi-Agent Systems, Tool Use & Planning, LLM Orchestration, Retrieval-Augmented Generation (RAG), LangGraph, LangChain, Prompt Engineering, Vector Search (pgvector), OpenAI API, Azure OpenAI (Familiar), Human-in-the-Loop Systems.

Languages: Python, JavaScript, SQL.

Databases: PostgreSQL, Supabase, MongoDB, Vector Databases.

Backend & APIs: FastAPI, Flask, Node.js (Express.js), REST APIs, Microservices, Async Services, State Machine Design.

Frontend: React.js, HTML/CSS, Responsive Design, UI/UX.

Auth & Security: OAuth 2.0, JWT, Session Management, Role-Based Access Control, Responsible AI Guardrails.

Cloud & Dev Tools: AWS, Git, Postman, Prisma ORM, n8n, VS Code.

CERTIFICATIONS

- Full-Stack Web Development — NullClass
- AWS Academy: Cloud Foundations & Machine Learning Foundations
- Salesforce Administrator Internship Certification

ADDITIONAL

Soft Skills: Problem Solving, Team Collaboration, Analytical Thinking, Continuous Learning

Languages: English (Professional), Telugu (Native), Hindi (Conversational)